

🔍 XAI: Input Gradients & Privacy Risks

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eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

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<https://interpretable-ml-class.github.io/>

Paper 1: Do Input Gradients Highlight Discriminative Features? ---

Do Input Gradients Highlight Discriminative Features?

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Abstract

Post-hoc gradient-based interpretability methods [1, 2] that provide instance-specific explanations of model predictions are often based on assumption (A): *magnitude of input gradients—gradients of logits with respect to input—noisily highlight discriminative task-relevant features*. In this work, we test the validity of assumption (A) using a three-pronged approach:

🍲 Do Attribution Methods Work? ---

Central Question

Do input gradient attribution methods actually highlight features that drive model predictions?

- Attribution methods assign **importance scores** to input features
- Common assumption: higher gradient magnitude $\Rightarrow$ higher feature contribution
- But... how do we *rigorously evaluate* this?

Challenge

Existing evaluations are often qualitative or rely on synthetic benchmarks — lacking a principled framework

Assumption (A) ---

Assumption (A): *Coordinates with larger input gradient magnitude are more relevant for model prediction compared to coordinates with smaller input gradient magnitude.*

Definition

Assumption (A): For model f and input x , feature i is more important than j if $|\nabla_{x_i} f(x)| > |\nabla_{x_j} f(x)|$

This assumption underlies most gradient-based XAI methods.

🍷 Key Contributions ---

DiffROAR: a new evaluation framework

Systematically evaluate whether attribution methods highlight truly discriminative features via retrain-and-compare

- **Multiple benchmarks:** SVHN, FashionMNIST, CIFAR-10, ImageNet-10
- **BlockMNIST:** controlled dataset isolating discriminative vs. spurious features
- **Feature Leakage Hypothesis:** theoretical explanation for when attribution *appears* to work
- **Theorem:** formal characterization of feature leakage conditions

(Shah et al. 2021)

Related Work ---

Sanity Checks (Adebayo et al.)

- Randomize model weights or labels
- Attribution must change accordingly

Fidelity-based Evaluation

- Mask top features, measure accuracy drop
- Single-pass evaluation can induce distribution shift

Adversarial Robustness

- Robust models can have more visually aligned gradients
- The paper asks whether they also satisfy Assumption (A)

Limitation of Prior Work

Prior tests do not directly compare top-attributed features against bottom-attributed features.

🍷 Why Sanity Checks Are Not Enough ---

Existing evaluation tools expose different failure modes:

- visual quality can be misleading
- randomization tests check sensitivity to weights/labels
- removal tests can create distribution shift
- ROAR retrains after removing important features

Gap

ROAR asks whether top-ranked features matter, but Assumption (A) is comparative: top-ranked features should matter *more than bottom-ranked features*.

(Adebayo et al. 2018; Hooker et al. 2019)

DiffROAR Framework ---

Setting ---

- Standard classification setting: each independently drawn data point is a pair $(x^{(i)}, y^{(i)})$ of instance and label.
- $x_j^{(i)}$ denotes the j th coordinate/feature of $x^{(i)}$.
- **Feature attribution scheme**

$$A : \mathbb{R}^d \rightarrow \{\sigma : \sigma \text{ is a permutation of } [d]\}$$

- maps a d -dimensional instance x to a permutation of its features.
- e.g., **input gradient attribution scheme** takes input instance x and predicted label $\hat{y}$, then ranks features in decreasing order of input-gradient magnitude.

$$A(x) : [d] \rightarrow [d]$$

🍷 Unmasking Schemes ---

- **Unmasked instance** x^S zeroes out all coordinates not in subset S .
- **Unmasking scheme** maps instance x to a subset $A(x)$ of coordinates.

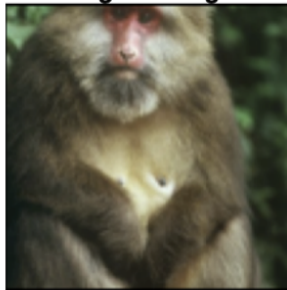
$$A : \mathbb{R}^d \rightarrow \{S : S \subseteq [d]\}$$

- **Top- k** : preserve the k coordinates with largest attribution scores.
- **Bottom- k** : preserve the k coordinates with smallest attribution scores.
- All other coordinates are set to zero.

DiffROAR intuition

If the attribution is faithful, top- k should be more predictive than bottom- k .

Original image



Top-k unmasked image



Predictive Power ---

For an unmasking scheme A , predictive power is

$$\text{PredPower}_M(A) = \sup_{f \in M} \mathbb{E}_{(x,y) \sim D} \left[\mathbf{1}\{f(x^{A(x)}) = y\} \right].$$

In plain English

How well can a model of architecture M classify examples if it only sees the features selected by A ?

- The model is retrained on the masked training set.
- Retraining reduces artifacts from showing an old model unusual masked inputs.
- The output is a test accuracy.

—

DiffROAR Metric ---

$$\text{DiffROAR}_M(A, k) = \text{PredPower}_M(A_k^{\text{top}}) - \text{PredPower}_M(A_k^{\text{bot}})$$

Interpretation of the metric:

- **Sign** indicates whether Assumption (A) is satisfied (> 0) or violated (< 0).
- **Magnitude** quantifies how strongly the attribution separates most and least discriminative coordinates.
- Random attribution should have expected DiffROAR near 0.

🍷 Experiment 1: Image Classification ---

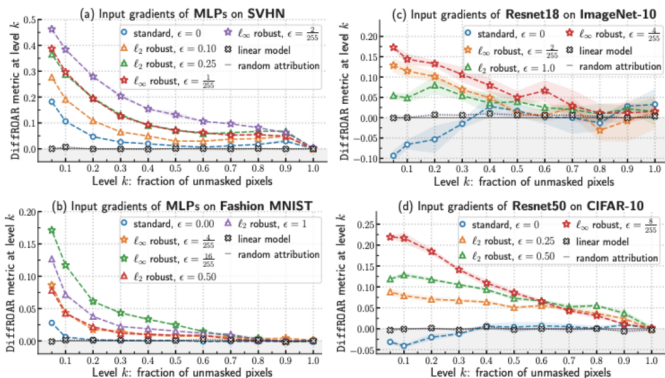
The first experiment tests Assumption (A) on standard visual datasets.

- **SVHN:** real-world house-number digits.
- **FashionMNIST:** clothing images.
- **CIFAR-10:** small natural images from 10 classes.
- **ImageNet-10:** reduced ImageNet with 10 super-classes.

Experimental Setup ---

- **Datasets:** SVHN, FashionMNIST, CIFAR-10, and ImageNet-10.
- **Models:** standard and adversarially trained two-hidden-layer MLPs and ResNets.
- **Adversarial training:** ℓ_2 and ℓ_∞ ϵ -robust models trained with PGD.
- **Metric:** DiffROAR across different unmasking fractions k .

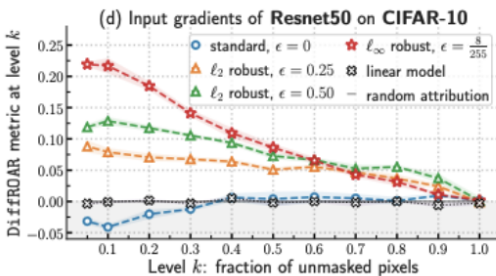
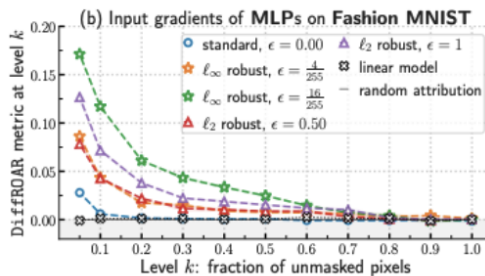
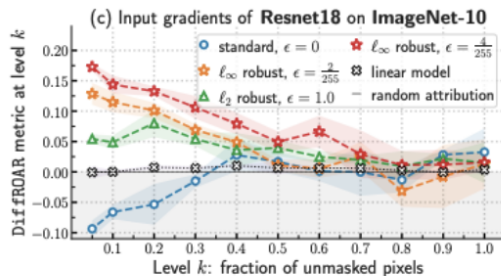
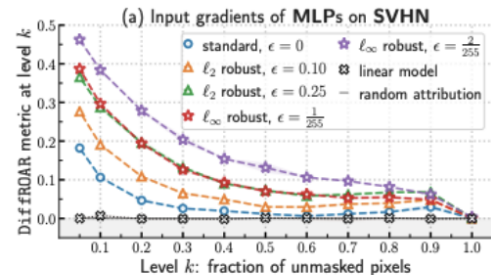
🍷 Results: Benchmark Datasets ---



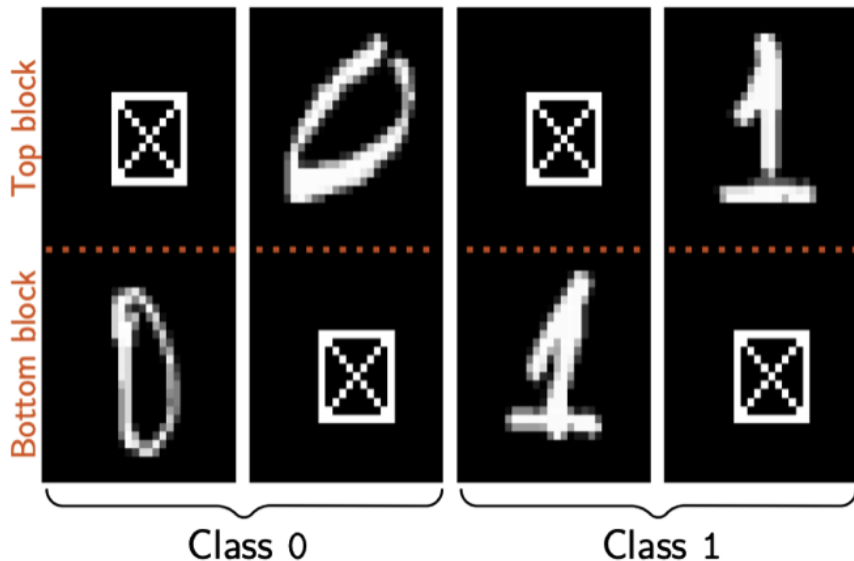
Surprising Finding

Most attribution methods yield $\text{DiffROAR} \approx 0$ on standard benchmarks — they perform no better than a **random feature selector**

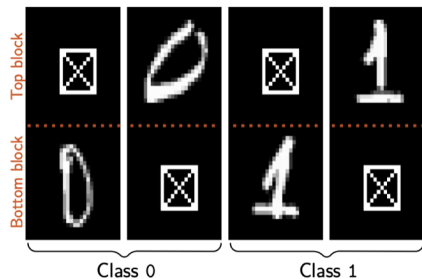
🍌 Results: Benchmark Datasets ---



🍷 Experiment 2: BlockMNIST ---



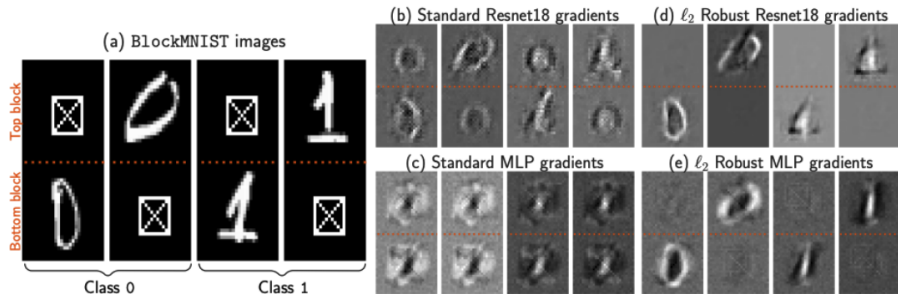
🍷 BlockMNIST Dataset ---



- Two MNIST digits placed **side by side**
- **Relevant digit**: correlated with class label
- **Spurious digit**: uncorrelated with label
- A good attribution method should highlight the **relevant** digit

🍷 Do Gradients Highlight the Signal Block? ---

🍷 Do Gradients Highlight the Signal Block?. Not Always! ---



Feature Leakage Hypothesis ---

Definition

Feature leakage: for a given instance, input gradients highlight not only the instance-specific signal, but also locations where signal appears in other instances.

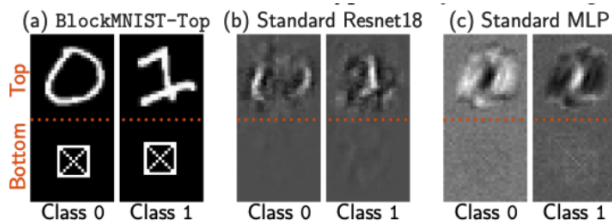
- In BlockMNIST, the signal block can appear at the **top** or the **bottom**.
- For the current image, only one block contains the digit that determines the label.
- Standard gradients may still highlight both locations, because both locations are discriminative somewhere in the dataset.

🍷 Testing Feature Leakage: BlockMNIST-Top ---

- Fix the signal block at the **top** for every instance.
- Keep the null block at the **bottom**.
- If leakage comes from variable signal location, standard gradients should improve.

Result

Standard ResNet18 and MLP now highlight the signal block and suppress the null block.



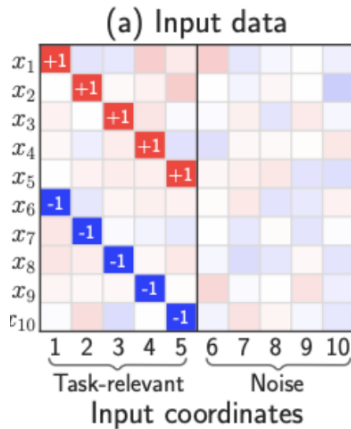
🍷 Simplified BlockMNIST: Theory Setup ---

To analyze feature leakage formally, the paper replaces images with a block-structured simplified dataset.

- Each input x is a concatenation of d blocks.
- The first $d/2$ blocks are **task-relevant**: the signal may appear in any one of them.
- For each instance, only one block $j^*(x)$ contains the actual signal for that example.
- The remaining $d/2$ blocks are **noise**: they never contain label information.

What a faithful gradient should do

Highlight the unique signal block $j^*(x)$, not all possible signal locations.



🍷 Theorem 1: Feature Leakage ---

Theorem 1. Consider distribution $\mathcal{D}$ (2) with $\eta < \frac{1}{10d}$. There exists a max-margin classifier ν^* for $\mathcal{D}$ in Wasserstein space (i.e., training both layers of FCN with $m \rightarrow \infty$) given by (4), such that for all $\forall (x, y) \sim \mathcal{D}$: (i) $\left\| (\nabla_x \mathcal{L}(\nu^*, (x, y)))_j \right\| = c > 0$ for every $j \in [d/2]$ and (ii) $\left\| (\nabla_x \mathcal{L}(\nu^*, (x, y)))_j \right\| = 0$ for every $j \in \{d/2 + 1, \dots, d\}$, where $(\nabla_x \mathcal{L}(\nu^*, (x, y)))_j$ denotes the j^{th} block of the input gradient $\nabla_x \mathcal{L}(\nu^*, (x, y))$.

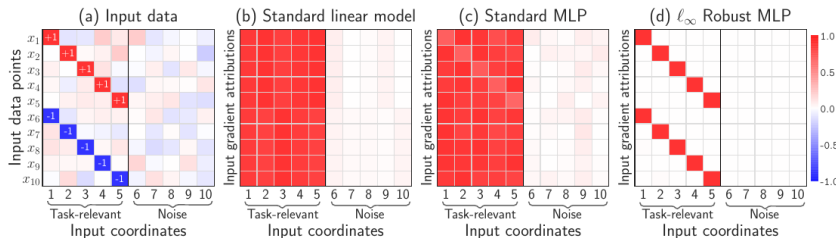
- There exists a max-margin classifier whose input-gradient magnitude is nonzero on all task-relevant blocks.
- But it is **constant across those blocks**: it does not isolate the unique signal block for the current instance.
- On noise blocks, the gradient is zero.

Meaning

This formalizes feature leakage: gradients highlight task-relevant locations that are not specific to the given instance.



Empirical Results: What to Look For ---



- All models get **100% test accuracy** on this simplified dataset.
- Linear model: marks all task-relevant coordinates.
- Standard MLP: still marks all possible signal locations.
- Robust MLP: marks the instance-specific signal coordinate.

Key point

Prediction can be perfect while the explanation is not instance-specific.

🍲 Conclusions: DiffROAR ---

Negative Result

Input gradient methods do **not** reliably highlight discriminative features on standard image benchmarks

DiffROAR ≈ 0 for most methods on SVHN, FashionMNIST, CIFAR-10, ImageNet-10

Key Nuance

Feature leakage explains apparent successes:

- BlockMNIST works due to spatial separation
- Not evidence of general semantic understanding

Discussion Questions: DiffROAR ---

- ① DiffROAR requires **retraining** — is this a realistic evaluation paradigm in practice?
- ② Does the feature leakage hypothesis generalize beyond synthetic datasets?
- ③ Can attribution methods be designed to be provably robust to feature leakage?
- ④ If attribution methods don't highlight discriminative features, what are they actually highlighting?

Paper 2: On the Privacy Risks of Algorithmic Recourse ---

ON THE PRIVACY RISKS OF ALGORITHMIC RECOURSE

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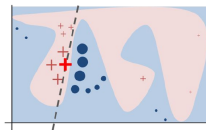
ABSTRACT

As predictive models are increasingly being employed to make consequential decisions, there is a growing emphasis on developing techniques that can provide algorithmic recourse to affected individuals. While such recourses can be immensely beneficial to affected individuals, potential adversaries could also exploit these recourses to compromise privacy. In this work, we make the first attempt at investigating if and how an adversary can leverage recourses to infer private information

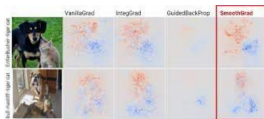
🍷 Privacy Risks in XAI ---

Central Question

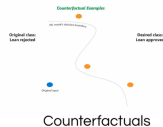
Can XAI mechanisms — specifically **algorithmic recourse** — leak private information about training data?



Lime



Gradient Based



- XAI outputs are information channels: they can reveal more than the intended explanation.
- This paper focuses on whether recourse can leak **training-set membership**.

Algorithmic Recourse ---

Context: a rejected loan applicant receives actionable steps to get approved

$$x' = \arg \min_{x' \in \mathcal{A}^P} \ell(f_\theta(x'), 1) + \lambda \cdot c(x, x')$$

- x' : counterfactual (the recourse)
- $c(x, x')$: cost/distance from x to x'
- $\mathcal{A}^P$: set of actionable features

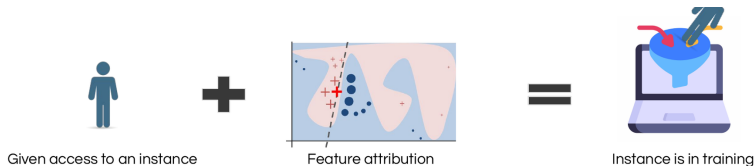
Definition

CFD (Counterfactual Distance): $c(x, x')$ — distance from input x to its recourse x'

Why CFD matters

The recourse algorithm outputs not only a counterfactual x' , but also a distance $c(x, x')$ that can be used as a statistic.

🍷 Previous Works ---



MI via Feature Attribution

- Shokri et al.: attribute $\rightarrow$ infer membership
- Requires **multiple queries**

Model Extraction via Counterfactuals

- Repeated counterfactual queries
- Also requires **many queries**

Limitation

Prior attacks assume multiple queries — impractical for real recourse systems

Novel Contribution

First **single-query** membership inference (MI) attack using counterfactual distances

- ① **CFD Thresholding Attack**: simple threshold on counterfactual distance
- ② **CFD LRT Attack**: likelihood ratio test using shadow models (more powerful)
- ③ **Theoretical bounds**: DP-based upper bounds on adversary success
- ④ **Empirical evaluation**: diverse datasets (Adult, HELOC, Diabetes) and recourse methods (SCFE, GS, CCHVAE)

(Pawelczyk et al. 2022)

Attack objective

Given a target instance, decide whether it belonged to the model owner's training set.

Loss Thresholding

- Training samples often have lower loss than test samples
- M_{Loss} predicts MEMBER when $\ell(f(x), y)$ is below a threshold

Loss LRT

- Uses shadow models to estimate loss/confidence distributions
- Predicts membership by comparing likelihoods
- More powerful than simple thresholding

Attack Assumptions ---

Info	Loss	CFD	Loss LRT	CFD LRT
Query access to f_{θ}	✓	×	✓	×
Query access to $\mathcal{R}$	×	✓	×	✓
Known loss function	✓	×	✓	×
Access to $\mathcal{D}^N$	×	×	✓	✓
Access to true labels	✓	×	✓	×

Important Distinction

Recourse-based attacks do not require true labels, the model loss, or direct query access to the predictive model.

- CFD attacks need access to the recourse system $\mathcal{R}$
- CFD LRT additionally assumes sample access to the data distribution $\mathcal{D}^N$

🍷 Recourse-based MI Game ---

Owner O (model provider):

- ① Trains f_θ on training set S
- ② Provides recourse $\mathcal{R}(x) \rightarrow x'$ when queried

Adversary $\mathcal{A}$ (attacker):

- ① Queries $\mathcal{R}(x)$ for target sample x
- ② Receives $\text{CFD} = c(x, x')$
- ③ Predicts: is $x \in S$?

Single-Query Constraint

The adversary can only query the recourse system **once** per sample — making this a realistic threat model

Intuition: Why CFD Works ---

Loss-based attacks are good at determining MEMBER, because model typically overfits to training points



This may be because, during training, decision boundary is pushed away from training points (Shroki et al.)



Points in training set should be further from boundary than points in test set



Counterfactual distance-based attacks!

Intuition

Training samples tend to be **further from the decision boundary** (the model fits them well). Their counterfactuals require a larger shift $\Rightarrow$ larger CFD.

MEMBER $\rightarrow$ large CFD (far from boundary)

NON-MEMBER $\rightarrow$ small CFD (closer to boundary)

🍷 Attack 1: CFD Thresholding ---

counterfactual distance between x , x' , from whichever recourse method

threshold

$$M_{\text{Distance}}(x) = \begin{cases} \text{MEMBER} & \text{if } c(x, x') \geq \tau_D \\ \text{NON-MEMBER} & \text{if } c(x, x') < \tau_D. \end{cases}$$

(recall: $M_{\text{Loss}}(x) = \begin{cases} \text{MEMBER} & \text{if } \ell(\theta, z) \leq \tau_L \\ \text{NON-MEMBER} & \text{if } \ell(\theta, z) > \tau_L. \end{cases} \right)$

$$M_{\text{Distance}}(x) = \begin{cases} \text{MEMBER} & \text{if } c(x, x') \geq \tau_D \\ \text{NON-MEMBER} & \text{otherwise} \end{cases}$$

- Simple and practical: only requires the recourse output $c(x, x')$
- τ_D chosen to maximize balanced accuracy at a given FPR

Attack 2: CFD LRT Intuition ---

LRT idea

Use counterfactual distance as the statistic, then ask whether the observed CFD looks likely under member vs. non-member behavior.

Full likelihood ratio

$$\Lambda = \frac{\Pr[c(x, x') \mid x \in D_t]}{\Pr[c(x, x') \mid x \notin D_t]}$$

- Inspired by loss-based LRT attacks
- Uses recourse distance instead of model loss

Practical one-sided version

- Exact member likelihood is computationally expensive
- Estimate the non-member CFD distribution using shadow models
- Test the observed CFD against a threshold for target FPR α

Attack 2: CFD LRT Algorithm ---

Algorithm 1 One-sided Distance-based Likelihood Ratio Test (CFD LRT)

```
1: Inputs: point  $(x, y)$ , recourse output  $s = \text{GetRecourse}(x, f_{\theta}), \mathcal{D}$ ; FP-Rate:  $\alpha$ , # Shadow Models:  $N$ ,  $\mathcal{T} = \text{TrainClassifier}(\cdot)$ 
2: teststats = []
3: Compute:  $t_0 = T(s) = c(x, x')$ 
4: for  $i = 1 : N$  do
5:   Sample  $\mathcal{D}_t^{(i)} \sim \mathcal{D}$ 
6:    $f_{\theta^{(i)}} = \text{TrainClassifier}(\mathcal{D}^{(i)})$ 
7:    $s^{(i)} = \text{GetRecourse}(x, f_{\theta^{(i)}})$ 
8:   teststats  $\leftarrow T(s^{(i)}) = c(x, x'^{(i)})$ 
9: end for
10:  $\hat{\mu}_{\text{MLE}} = \frac{1}{N} \sum_{i=1}^N (\log c(x, \mathbf{x}'^{(i)}))$ 
11:  $\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{N} \sum_{i=1}^N (\hat{\mu}_{\text{MLE}} - \log(c(x, \mathbf{x}'^{(i)})))^2$ 
12: if  $t_0 > z_{1-\alpha}$  then
13:   Output:  $G = \text{NON-MEMBER}$ 
14: else
15:   Output:  $G = \text{MEMBER}$ 
16: end if
```

▷ $z_{1-\alpha}$ is the $1-\alpha$ -quantile of $Z \sim \mathcal{LN}(\hat{\mu}_{\text{MLE}}, \hat{\sigma}_{\text{MLE}}^2)$

Advantage

Leverages distributional information from shadow models — more powerful than simple thresholding in higher-dimensional settings

🍷 Privacy Bounds via Differential Privacy ---

Theorem 1

If the model training mechanism is ε -DP, then for any MI adversary $\mathcal{A}$:

$$\text{BA}_{\mathcal{A}} \leq \frac{1}{2} + \frac{1 - e^{-\varepsilon}}{2}$$

DP is Not a Silver Bullet

Small ε bounds adversary success — but significantly degrades recourse quality (utility-privacy tradeoff)

Experimental Setup ---

Datasets:

- Adult (income prediction)
- HELOC (credit risk)
- Diabetes
- Synthetic (controlled)

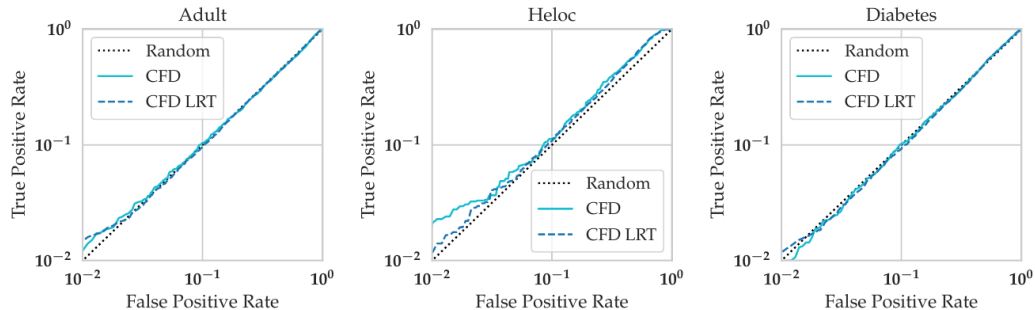
Recourse Methods:

- SCFE
- GS (Growing Spheres)
- CCHVAE

Baselines: M_{Loss} threshold, Loss LRT



Attack Efficiency: ROC Curves (SCFE) ---

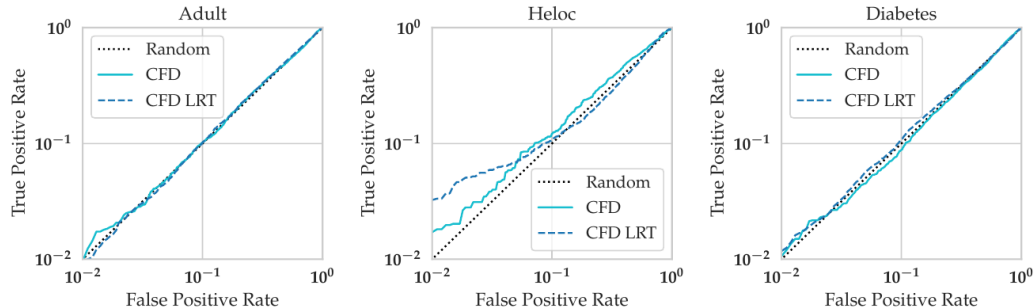


(a) SCFE

How to read

Curves above the random diagonal indicate membership signal at that false-positive rate.

🍷 Attack Efficiency: ROC Curves (GS) ---

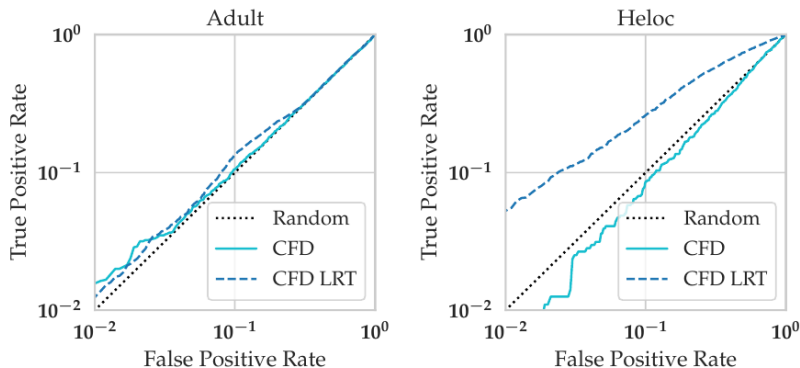


(b) GS

How to read

The low-FPR region is especially important: even a small high-confidence leakage can be privacy-relevant.

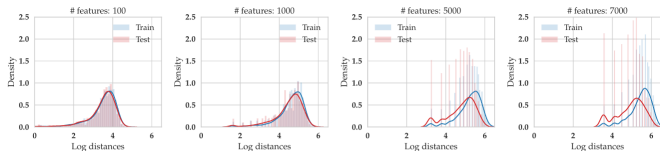
🍷 Attack Efficiency: ROC Curves (CCHVAE) ---



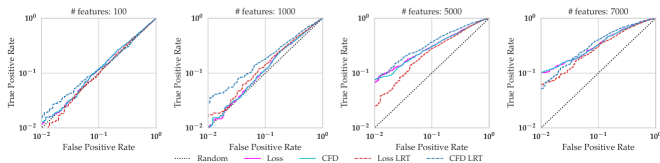
Key Takeaway

CFD LRT is often above the random baseline, especially on HELOC — counterfactual distances can carry membership signal.

🍷 Effect of Feature Dimensionality ---



(a) Comparing log distances (ℓ_1) to the decision boundary across train and test points.

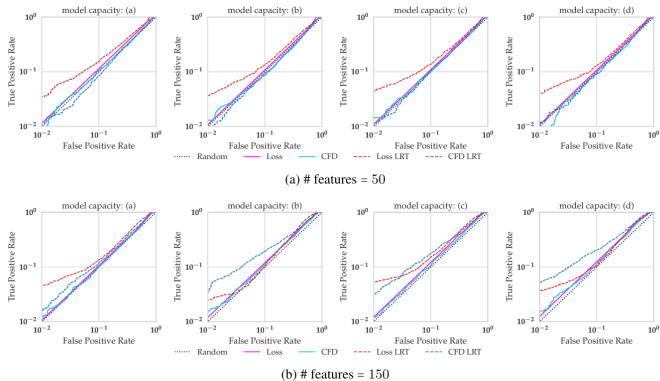


Finding

Higher-dimensional feature spaces $\Rightarrow$ greater MI attack success

More features $\rightarrow$ richer counterfactual signal $\rightarrow$ more distinguishable CFD distributions

Effect of Model Architecture ---



Finding

More complex model architectures $\Rightarrow$ greater MI attack success

Complex models memorize training data more strongly $\rightarrow$ larger margin for training samples $\rightarrow$ larger CFD gap

Conclusion: Privacy Risks ---

Novel Attacks

- Leverage recourses to infer training data membership
- MI attacks using counterfactual distances (**CFD**)
- **Single-query** — practical threat model

Evidence of Privacy Leakage

- Recourse algorithms carry membership signal
- **Explainability-privacy tradeoff** is real
- Attacks effective across diverse domains (lending, healthcare, law)

Limitations ---

- CFD is only a **heuristic** — an approximation of distance to the decision boundary
- Single-query assumption (adversary can only query once per sample)
- Must assume adversary knows the optimal threshold maximizing TPR at fixed FPR
- Paper highlights the **problem**, not yet a **solution**
- Evaluated on binary classification tasks only

Future Work ---

- **Generalization**: can recourse lead to reconstruction attacks or attacks on training data statistics?
- **Other XAI mechanisms**: which other explanation methods involve privacy violations?
- **Solutions to protect privacy**: train models that provide recourse while mitigating privacy risks
 - ▶ How to construct faithful explanations that don't leak training data?
 - ▶ What is the privacy-utility trade-off?

(Pawelczyk et al. 2022)

Discussion Questions: Privacy Risks ---

- ➊ Given the explainability-privacy tradeoff, what is the role of **ML practitioners** vs. **end users** in setting explainability and privacy benchmarks?
- ➋ Both privacy and explainability cultivate user *trust* in ML models. In what situations would you prioritize one pillar over the other?
- ➌ Besides recourse, what other XAI mechanisms might lead to privacy violations?
- ➍ Is it even possible to have **private explanations**? Is this worth pursuing?

Thank You!

References --- |

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